MANTA

A wave that rose from the sea floor and stayed.

One ribbon, cut like a baguette so it prints at home and still holds an adult.



THE STORY

Once, a wave rose from the sea floor and did not fall back. Its feet still hold the seabed like the fins of a manta; its body swells into a seat, climbs into a backrest and breaks at the crest, curling forward and falling back *through* itself to the floor — one ribbon, no separate legs, no frame. The colour follows the water: deep blue at the feet, turquoise through the seat, pale foam at the crest. Every seam is left white — epoxy thickened with silica, foamed like the spray between waves. Seen from the front, the seat writes an **M**: for Manta, and for *Mare*, the sea that makes most of the oxygen we breathe. Flutes follow the ribbon's motion, bold at the floor, calming toward the crest, smooth where you sit.

BIOMIMICRY AS STRUCTURE, NOT ORNAMENT

- **Wolff's law** — thickness follows the bending moment: 84 mm at the lumbar knot, 32 mm at the crest.
- **Murray's law** — the lumbar, the one true junction, is the cube-root-sum of its branches.
- **Spiral fibre** — flutes follow the stress path and stiffen by corrugation; the section twists up to 20°.
- **Shell** — the closed section and dished seat carry load as a shell.
- **Fiddlehead scroll** — the crest curls back through the backrest: a closed loop, not a cantilever.
- **Buttress roots** — each foot forks into two splayed prongs.

BUILT, AND PRINTABLE AT HOME

41 segments (each ≤ 212 × 220 × 250 mm), auto-oriented: under 1 % of the filament is support (PrusaSlicer), none inside the holes. PETG, 0.2 mm layers, 35 % gyroid: ~16.5 kg, ~460 h on one printer, ~90 € filament + ~20 € epoxy. **A full-size prototype is printed and assembled** (the light variant, 75 % wall thickness) — photos on the board.

ASSEMBLY — COMPUTED JOINTS, NO SCREWS

The script slices the ribbon crosswise by length and curvature, keeping every cut ≥ 70 mm from the knee and lumbar, where bending peaks. Every joint is a full solid cross-section (7,000–17,000 mm²), epoxied, with a self-centring conical peg sized to the local thickness; the two most-loaded joints add a solid Ø11 pin. Where the crest returns through the backrest it is trimmed to the backrest's shape and glued on as a saddle. Part numbers are engraved on the glue faces. One command (manta_verify.py) re-checks everything: strength, a **virtual assembly** (pegs seated, 2 mm walls, no overlaps), bed fit, real-slicer support.

STABILITY AND LOAD

First-order check (120 kg × 1.8 dynamic × 2.0 safety): every member passes; glued full-section joint 26×, weakest glued check 2.1×; with no glue at all, peg and pin still carry 0.9×. Tipping 0.56 sideways, 0.55–0.77 rearward; safe static load ~250 kg; 8/8 ergonomics checks pass.

~250 kg	26×	~16.5 kg	~110 €
safe static load	glued full-section joint	finished weight	filament + epoxy

OPEN DESIGN — NOTHING REPEATS

Nothing in nature repeats, so no two of the 41 segments do either. Change the tables and the script regenerates a leaner chair with its joints and checks; the 70 kg variant built here is in the repo.